

AI News Intelligence Agent

Problem: Staying current on AI research and industry news means manually checking a dozen scattered sources — company blogs, YouTube channels, papers every day. It's slow and easy to fall behind on.

Solution: A website that runs a scheduled agentic pipeline (LangGraph) which collects new content from RSS feeds, cleans it, summarizes and classifies it with an LLM, compiles a digest, and emails it while also serving that same digest as a live, browsable website.

Core value: open the site, see today's AI news already read and organized for you, with links back to every original source.

Project Overview

- Built an autonomous AI-powered news research system.
- Collects the latest AI news from trusted RSS sources.
- Extracts complete article content.
- Uses an LLM to summarize and categorize news.
- Stores processed data in SQLite.
- Displays news through an interactive Streamlit dashboard.

Technology Stack

- **Python** – Core programming language
- **Streamlit** – Dashboard frontend
- **SQLite** – Database
- **SQLAlchemy** – Database ORM
- **LangChain** – LLM integration
- **LangGraph** – AI workflow orchestration
- **Groq API** – Fast LLM inference
- **Feedparser** – RSS feed collection
- **BeautifulSoup** – HTML parsing
- **Newspaper3k** – Article extraction
- **GitHub** – Version control
- **Streamlit Cloud** – Deployment

Project Folder Structure

AI News/

```
|
|
|— app/
|— collectors/
|— config/
|— db/
|— pipeline/
|— services/
|— utils/
|— requirements.txt
|— README.md
|— ai_digest.db
└— .env
```

Important Files

app/

- dashboard.py — Streamlit dashboard

collectors/

- blog_rss.py — Collect RSS news
- article_scraper.py — Extract article content

services/

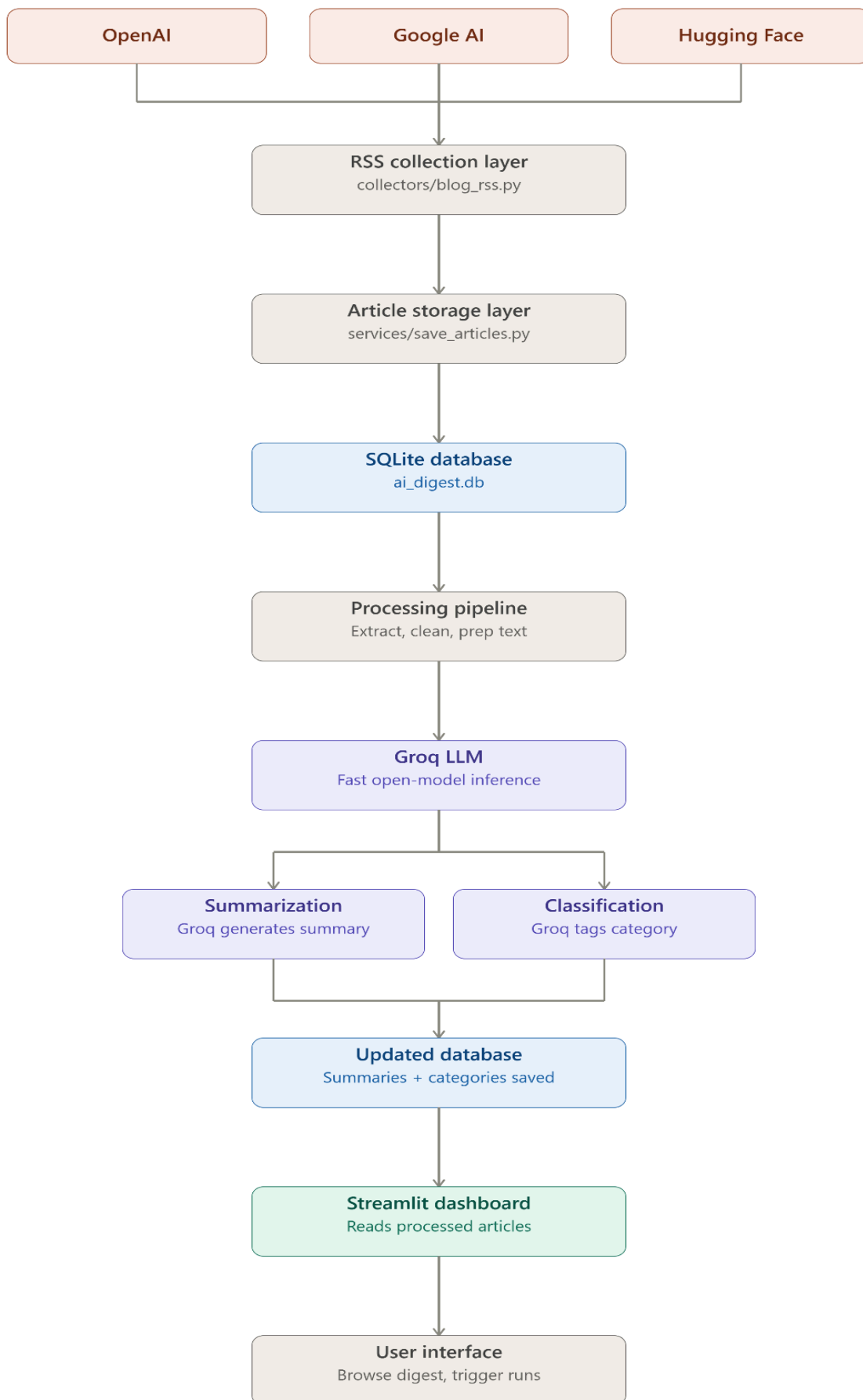
- save_articles.py — Save articles to database
- summarizer.py — Generate AI summaries
- process_articles.py — Process and categorize articles

db/

- database.py — Database connection
- models.py — Database models
- init_db.py — Create database tables

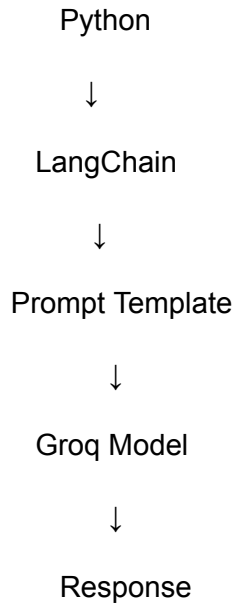
config/

- sources.yaml — RSS feed configuration



LangChain Role

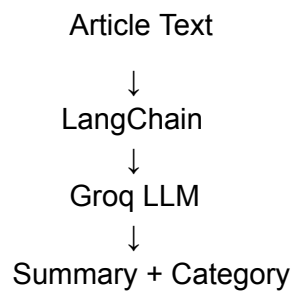
LangChain is the framework connecting your application with the LLM.



LangChain handles:

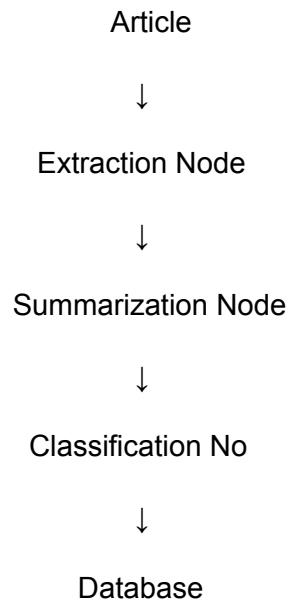
- Model connection
- Prompt management
- Output processing

AI Processing



LangGraph Role

LangGraph manages AI workflow.



Conclusion

- Built an end-to-end AI news automation system.
- Integrated RSS collection with AI processing.
- Applied LLMs for summarization and categorization.
- Developed a responsive Streamlit dashboard.
- Deployed the project on GitHub and Streamlit Cloud.
- Demonstrated practical skills in Python, AI, databases, and web deployment.